

State of Multi Hazard Early Warning Systems in Africa

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Early Warnings for All (EW4All) is a groundbreaking initiative that aims to protect everyone everywhere by 2027



- Announced by UN Secretary General Antonio Guterres on **23 March 2022** (World Meteorological Day), **launched at COP27** that year as part of **Acceleration Agenda**
- Aligns with the priorities of the **Paris Agreement**
- Supports key provisions of the **Sendai Framework for Disaster Risk Reduction 2015-2030**, particularly **Target G** on availability and accessibility of multi-hazard early warning systems
- Contributes directly to the UN's **Sustainable Development Goals** (SDGs), on poverty, hunger, health, water, clean energy, climate action and sustainable cities
- **NOT a new, standalone programme or project!**
→ EW4All builds on existing or previous initiatives and partnerships



Pillar 1:
Disaster Risk Knowledge



Ensuring all countries have access to reliable, understandable and relevant risk information, science and expertise

Pillar 2:
Detection, observation, monitoring, analysis, forecasting of hazards



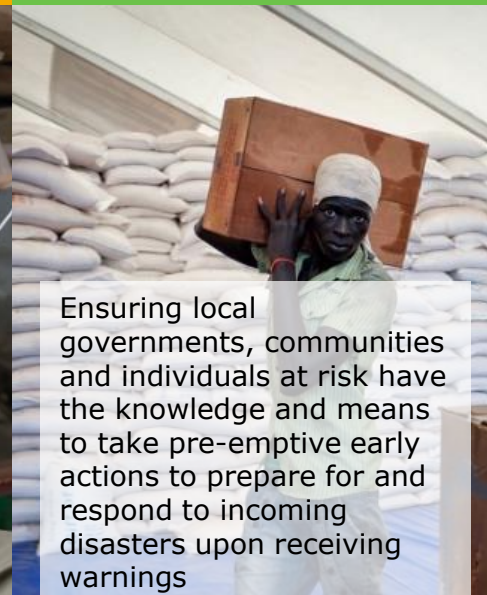
Ensuring all countries have robust forecast and monitoring systems, enabling policies to support optimization and sustainability of hazard monitoring and early warning systems

Pillar 3:
Warning Dissemination & Communication



Using a people-centred approach to ensure that early warnings are effectively and timely disseminated to reach everyone, especially those most at risk

Pillar 4:
Preparedness & Response Capabilities



Ensuring local governments, communities and individuals at risk have the knowledge and means to take pre-emptive early actions to prepare for and respond to incoming disasters upon receiving warnings



Built on four Pillars with Inter-Pillar Domains

1. **GOVERNANCE**, policy & legislation
2. Stakeholder coordination & **PARTNERSHIPS**
3. **Advocacy** & awareness
4. Planning & **FINANCE**
5. Monitoring & Evaluation (**M&E**)



MHEWS coverage in Africa

More countries are developing, implementing and improving their MHEWS, but critical gaps remain, especially in Least Developed Countries.

Additionally, early warning systems coverage in Africa is uneven and insufficient, especially in fragile and conflict-affected areas.

Progress in implementing EW4All in Africa

About 20 countries report to have MHEWS as of 2025

Global Hazard Detection is steadily improving in Africa

Huge gaps still exist

33 Africa Countries report dissemination of warnings using various means

For example, 24 report use of localized SMS

Developing Anticipatory Frameworks to address response

2025 – at least 20 countries



Continued efforts to
implement the
Action Plan

- Identified activities
including data sharing

Demonstrating Results towards implementation of EW4All in Africa

Expanded EW4All geographical coverage in the region

- EW4All workshops conducted in 30 African countries
- A total of 21 roadmaps finalized with 9 under review
- In 15 countries, that have indicated their interest to join the initiative, preparatory works are ongoing

Strengthened legal instruments to improve effectiveness in early warning/anticipatory actions

- WMO is supported countries in improving their legal frameworks
- Ongoing processes in some countries

Strengthened forecasting and computing capacities across time scales

- Improved access to regional and global Numerical Weather Prediction
- Severe Weather Forecasting capabilities improved
- Drought and Flood systems improved

implementation of EW4All Africa

Enhanced Regional and National Capacities in Alert Communication and Dissemination

- Strengthened institutional capacity in CAP implementation with use of the CAP standard increased by more than 100% in 41 African countries.

Enhanced EWS through Simulations and After Event Reviews

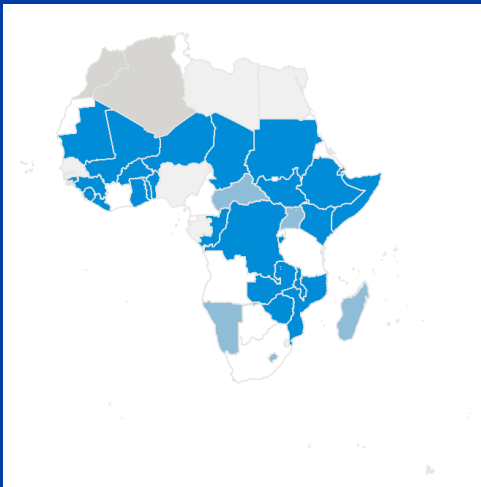
Webinars on Early Warning and Early Action

- Least Developed Countries in Conflict and Post-conflict Situations in Africa
- UN agencies
- National Webinars
- Accelerating Cell Broadcast

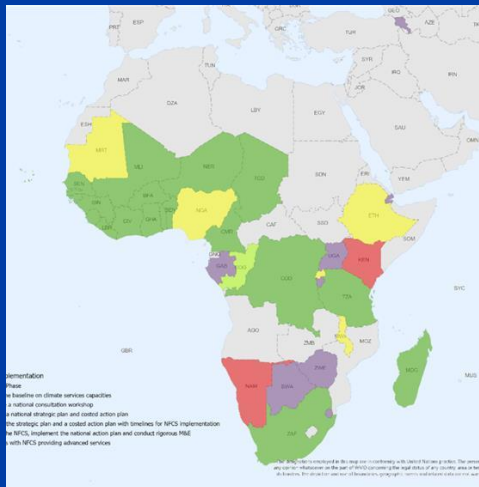
Enabled AA to reduce disasters impacts through Situation Rooms – **still existing some challenges**

Strengthening Community Early Warning Systems

ClimWeb



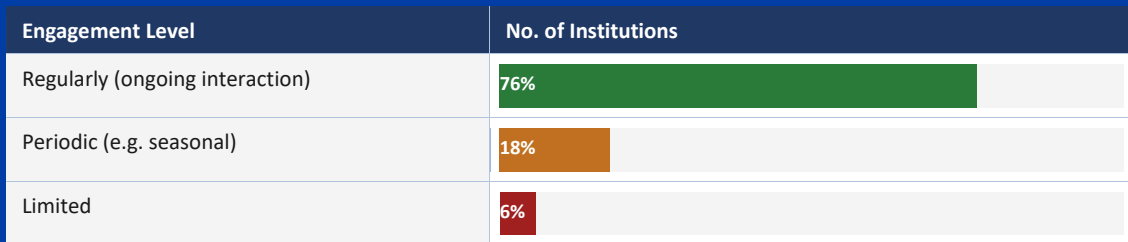
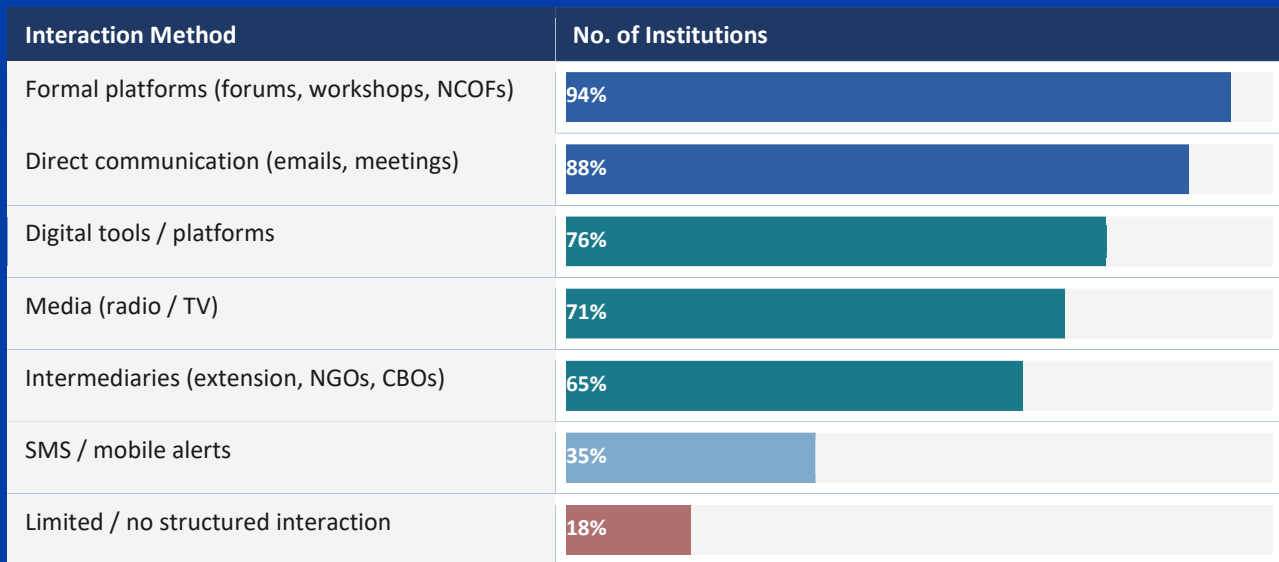
NFCS



EW4ALL



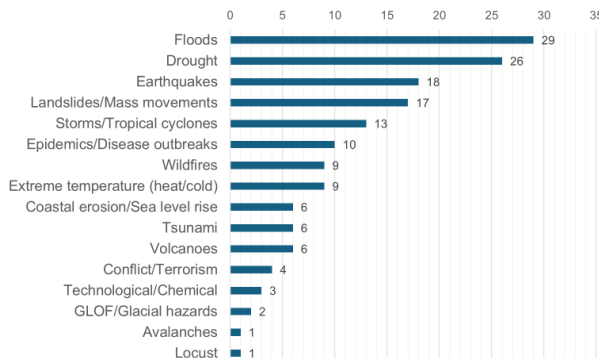
Current Interaction dissemination of EWS



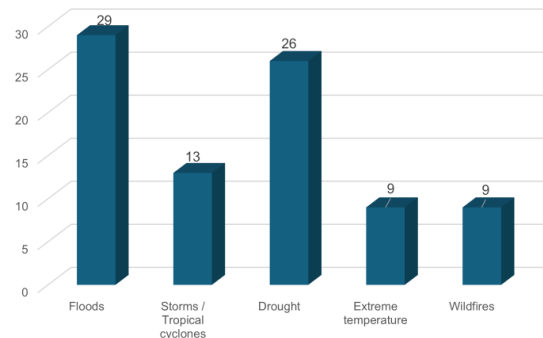
Roadmap review – Top Priority Hazards

Top 5 Hazards

Frequency of overall hazards in 31 countries



Frequency of TOP 5 hazards in 31 countries



Need to formulate required intervention

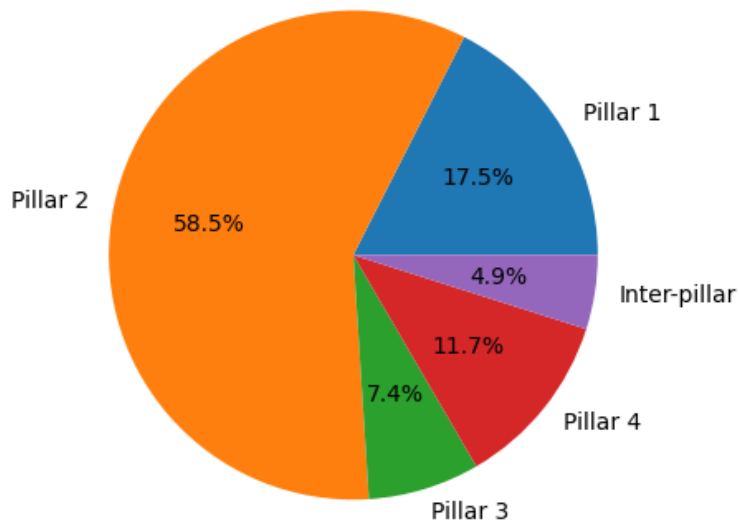
Budget Allocation in 5 Pillars in sample EW4All Roadmaps

16/31 countries have clarified in the roadmap the national budget allocation across all pillars.

Country list

- Gambia
- Kenya
- Seychelles
- South Sudan
- Tanzania
- Uganda
- Lao PDR
- Nepal
- Guyana
- Burundi
- Madagascar
- Haiti
- Chad
- Ecuador
- Guatemala
- Mozambique

Total Budget Distribution Across 16 Countries

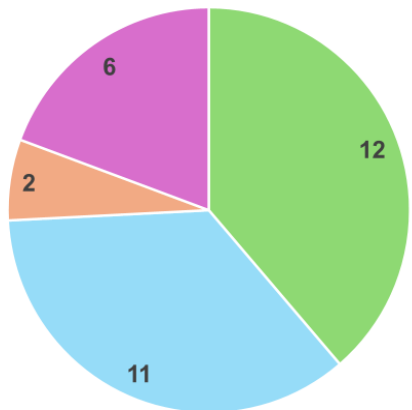


Pillar 2 - heavy

- Infrastructure-led (weather stations, radar, observations networks etc)
- Data exchange & system building
- Forecasting capability uplift
- Technical training etc

Inter-pillar coordination mechanism

Inter-pillar coordination mechanism & dedicated budget



- coordination mechanism + budget
- coordination mechanism + budget not specified
- unclear mechanism + budget
- unclear mechanism + budget not specified

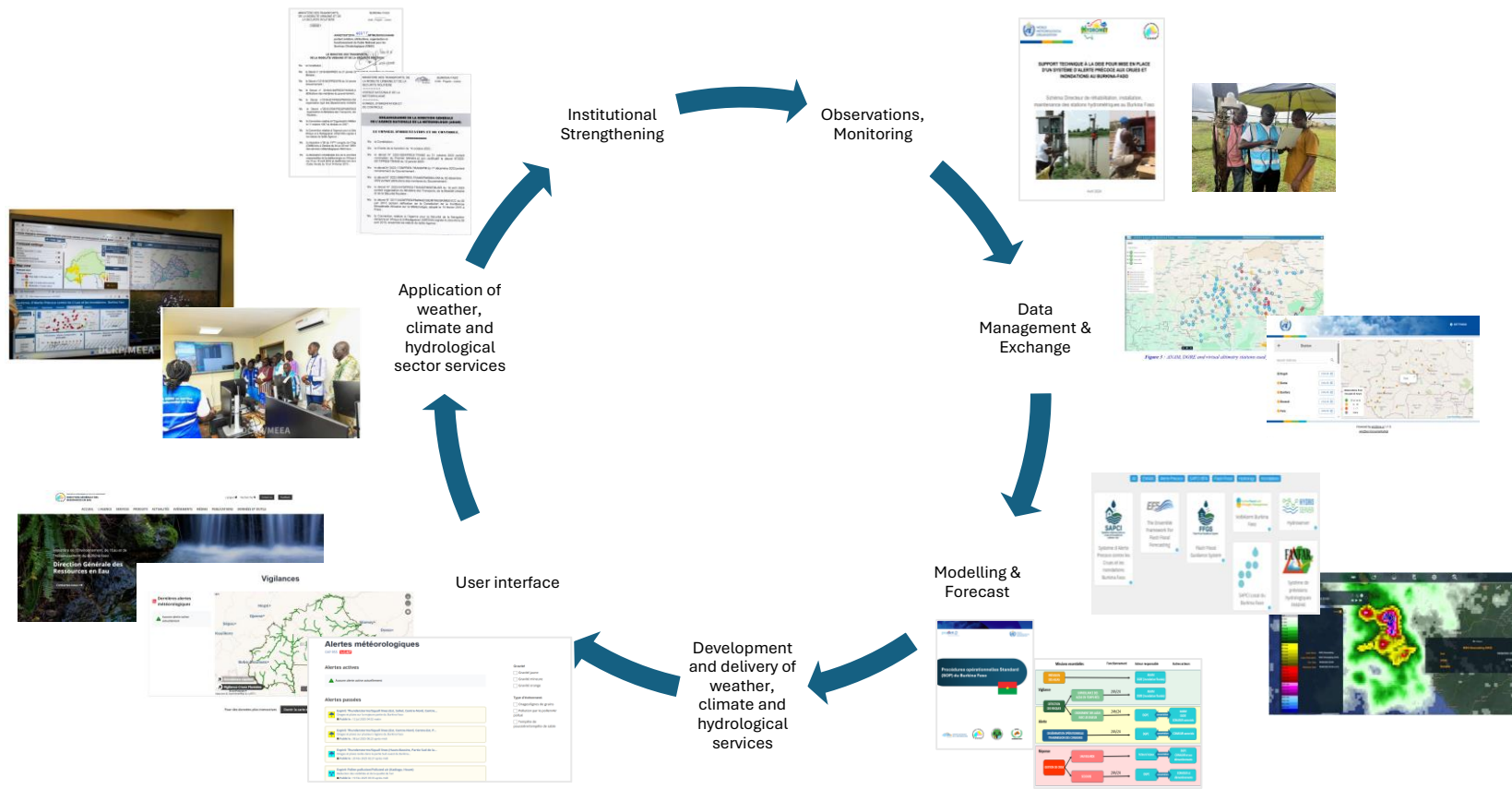


WORLD
METEOROLOGICAL
ORGANIZATION

Clear inter-pillar coordination mechanism

- **Burundi:** “Coordination anchored; governance structure shows multi-level platform (national→provincial→communal→community) and inter-pillar coordination, backed by Presidential Decrees establishing the platform.”
- **Ethiopia:** “Oversight by a National DRM Council; EDRMC as coordination lead; multi-level platforms with three task forces (Technical, M&E/Reporting, Policy).”
- **Gambia:** “Establishes an ‘EW4All Coordination Committee’ with defined functions (meetings, resource mobilization, M&E, legal/policy harmonization); chaired rotationally by pillar leads (initial chair DWR).”

Technical Support to Members



Various Assessments in Support to Members

Informing policy and investment decisions for high-quality weather forecasts, early warning systems, and climate information in developing countries

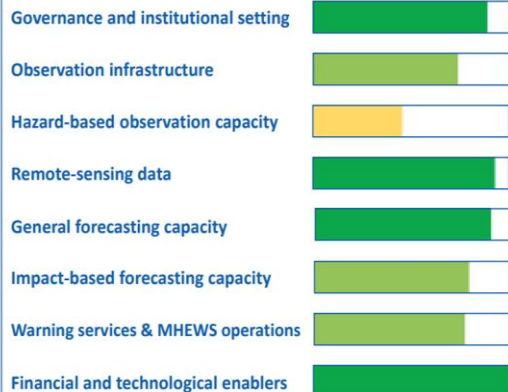
ETHIOPIA

4

The NMHS has a strong mandate, good inter-agency cooperation, an extensive and well-maintained observing infrastructure network (incl. radars) and financial resources for providing comprehensive forecasting and warning services. Having piloted its IBF potential readiness in certain areas (drought, agriculture), the NMHS requires knowledge and technology transfer to scale up its IBF and MHEWS against hydromet hazards (floods, landslides).



Pillar II capacity



The capacity assessment level above is ranked on a scale from 1 to 5, where 5 represents advanced capacity and 1 represents no capacity. The capacity level is determined via a quantitative (weighted rating) and qualitative analysis of the EW4All Rapid Assessment Tool (RAT) submission of the Member.

The colours of the bars above express the degree of attainment of each element as quantified by the EW4All RAT, following the percentages indicated hereafter.



Drought/dry spell

+ Good monitoring, forecasting and warning capacity, incl. IBF — Partial lack of soil moisture data

Flash-floods

+ Good monitoring, forecasting and warning capacity, incl. IBF

Riverine floods

+ Good monitoring, forecasting and warning capacity, incl. IBF

Landslides/mudslides

— Observations (local and remote) and risk maps are missing to complete monitoring and IBF capacity

Wild land fires/forest fires

— Observations (local and remote) and risk maps are missing to complete monitoring and IBF capacity

Country Hydromet Diagnostics

Components

	Indicators	Maturity level	Data sources
10 critical elements of the hydromet value cycle	Standardized indicators for each critical element	Informing where additional focus and support is needed	Peer assessment based on a data sources tiered-approach: Self-assessed Objective data Client survey-based

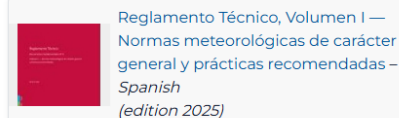
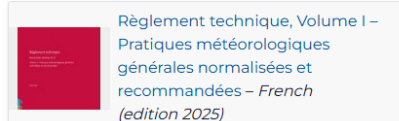
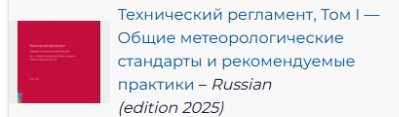
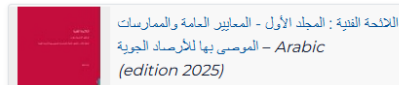
- **Climate Services Checklist**
- **Survey on Hydrological Capabilities**
- **Peer review processes.**
- **Data sharing**

Updated Early Warning Technical Regulations

- Agreed by WMO Extraordinary Congress October 2025, now published in six official languages
- Substantial upgrade to agreed standards of early warning services to be implemented, including:
 - Definition of institutional arrangements
 - Coordination between entities
 - Quality and business continuity of services received by public
 - Information about possible impacts of hazards on communities, infrastructure and relevant socioeconomic sectors
 - Communication standards and channels
 - Recording and cataloguing of events and impacts
- For 'meteorological, hydrological, climatological, and related environmental hazards'



*WMO-No.49, Part IV, 6:
Early Warning Services*





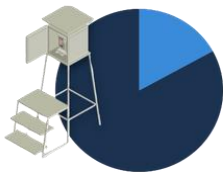
Systematic Observations Financing Facility

Weather
and climate
data for
resilience

SOFF supports countries with the most severe observation gaps, prioritizing Least Developed Countries (LDC) and Small Island Developing States (SIDS).

Africa

12%
GBON
compliant
surface
stations



12%
GBON
compliant
upper air
stations

25 countries

Burkina Faso, Cabo Verde, Chad, Comoros, Democratic Republic of the Congo, Djibouti, Ethiopia, Guinea, Guinea Bissau, Liberia, Madagascar, Malawi, Mauritius, Mozambique, Niger, Rwanda, Sao Tome and Principe, Senegal, Seychelles, Sierra Leone, Somalia, South Sudan, Uganda, United Republic of Tanzania, Zambia

USD 71 million Budget approved

SOFF Readiness support:

- **25 countries** benefitting
- Approved budget: **USD 3.7 million**

SOFF Investment support:

- **13 countries** approved budget: **USD 67.5 million**
- **1 country** (*Senegal*) in the Investment pipeline

SOFF Implementation Partners

Peer advisors: Austria, Denmark, Finland, Germany, Iceland, India, Morocco, Netherland, Nigeria, Norway, Portugal, South Africa, Spain, Switzerland, United Kingdom

Implementing Entities: AfDB, WFP, FAO, UNDP, World Bank, IsDB, UNEP

Education and Training opportunities supported by WMO in RA-I (2021 – 2025)



Specialized short-term courses delivered online and in person, covering meteorology, climatology, agrometeorology, hydrology.



technician qualification programs including BIP-M, BIP-MT, entry-level and senior technician training, meteorological engineering training, foundation training and on-the-job training for operational meteorologists and technicians.



formal degree pathways in meteorology, agrometeorology, climatology, hydrology, hydrometeorology (BSc, MSc and PhD programs)

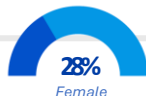
Fellowships

116 awarded

to RA I Members

28

Members

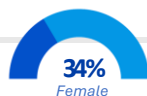


Training

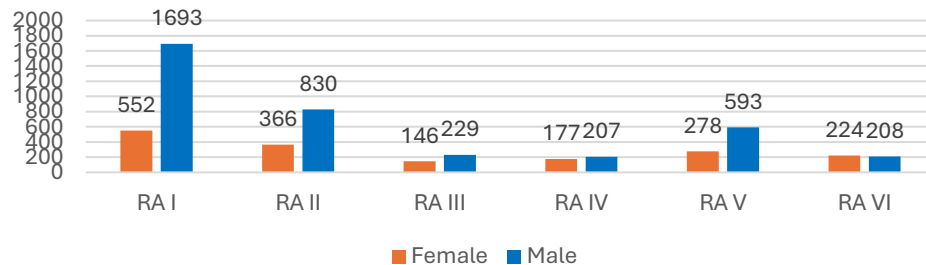
10,367 Participants

10 RTC Components

469 Courses Delivered



WMO Supported Training Participants (2021-2025)



Between 2021 and 2025, WMO supported 2'245 participants from African Union Member States in 99 courses hosted by WMO and in 37 Institutions, 552 females (24.6%) and 1'693 males (75.4%).

Africa Early Warnings Forum, Addis Ababa, Ethiopia 2-4 September 2026

Organizer(s) United Nations Office for Disaster Risk Reduction - Regional Office for Africa, African Union Commission, Meteorological Organization (WMO), International Telecommunication Union (ITU), International Federation of Red Crescent Societies (IFRC), Ethiopia - government



Bringing together representatives from a wide range of stakeholders - --- the forum took stock of progress, identified operational priorities, and mobilizing of financing and innovation needed to accelerate implementation by and beyond 2027.

Aimed at addressing the increasing exposure to climate change and its associated costs, both globally and particularly in Africa.

The summit culminated in the Nairobi Declaration, an eleven-point call to action proclaiming African States' unified stance on climate action.



The Addis Ababa Climate Summit (ACS2), September 8-10, 2025, significant climate diplomacy.

Supported acceleration of global climate solutions and financing for Africa.

AFRICA'S 1ST MULTI STAKEHOLDER FORUM ON EARLY WARNINGS FOR ALL FORUM' HOSTED BY NAMIBIA

Key Issues and Challenges

- Limited resource mobilization: Need scalable domestic financing and public–private partnerships.
- Ownership: Insufficient country-led action slows coordination and sustainability.
- Capacity: Gaps across the value chain—from data to impact-based services to community response.
- Underutilization of regional organizations to accelerate progress.
- Fragile contexts: Need tailored approaches and basic functionality (incl. situation rooms) in FCV settings.
- Data fragmentation and need for increased sharing



Challenges of implementing EW4ALL related to Pillar 2

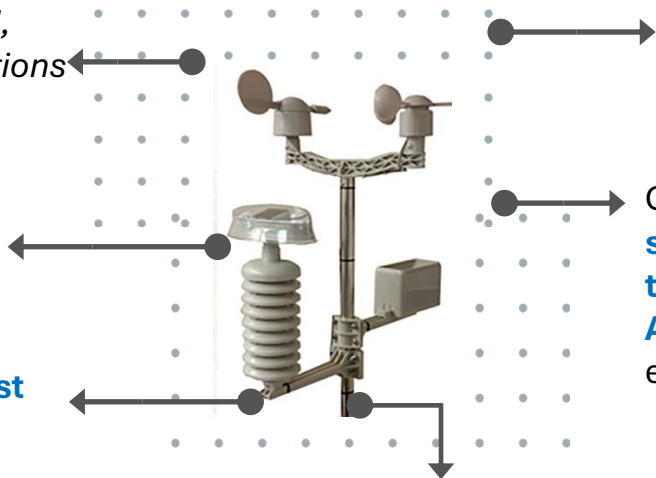
Integrated Observing Systems –

need to strengthen WIGOS, GBON, RBON, AMDAR, upper air observations and uptake of WMO Unified Data Policy

Coordinated Regional Centers –
RCCs, RSMCs, RTCs, RICs, WIGOS Centers

Implementation of **Weather Forecast Competency Frameworks** to strengthen staff capacity

WMO is addressing these gaps through initiatives such as : **SOFF, CREWS, SWFP, QMS, Digital Transformation**



National capacity for accurate and timely warnings: (Plans, strategies, legislation, human and institutional resources).

Capacity of **Members in the use of satellite observations and advanced technologies (radar, lightning detection, Artificial Intelligence)** and other emerging technologies

Improved data quality through; data rescue, digitization, visualization, automation of the data collection process, quality control, archiving, and retrieval of data in the NMHSs DMS.

End